

Sri Lanka Institute of Information Technology



SE2030 – Software Engineering
Year 2, Semester 1- 2026

Web-based Diet Planning & Health Check-up System

Lab 02 Report

2026-Y2-S1-MLB-B5G1-01

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Scrum Roles

Product Owner: Gunathilaka M.D.S.T (IT25100792)

Scrum Master: Punsara M. H. (IT25103601)

Developers:

- **Vidanage.T.L (IT25101803)**
- **Herath H.M.H.Y (IT25102636)**
- **Hasaranga S.O (IT25103681)**
- **Mohanaranjan M. (IT25101696)**

Identified Personas & User Stories

1. Persona: Patient

- As a Patient, I want to register an account so that I can access healthcare services online.
- As a Patient, I want to book and manage appointments so that I can consult doctors and dietitians conveniently.
- As a Patient, I want to view my health check-up results so that I can monitor my health condition.
- As a Patient, I want to access my diet plan and progress records so that I can follow my nutritional goals effectively.

2. Persona: Dietitian

- As a Dietitian, I want to create personalized diet plans so that patients can achieve their health goals.
- As a Dietitian, I want to update diet plans so that recommendations remain suitable for patient progress.
- As a Dietitian, I want to view patient health records so that I can provide accurate dietary advice.
- As a Dietitian, I want to communicate with patients so that I can provide guidance and answer queries.

3. Persona: Doctor

- As a Doctor, I want to access patient medical records so that I can make informed decisions during consultations.
- As a Doctor, I want to record health check-up results so that patient information stays up to date.
- As a Doctor, I want to review patients' health progress so that I can recommend appropriate treatments.

- As a Doctor, I want to schedule follow-up consultations so that patients receive continuous care.

4. Persona: Reception Staff

- As a Reception Staff Member, I want to register new patients so that their information is available in the system.
- As a Reception Staff Member, I want to manage appointments so that schedules remain organized and conflict-free.
- As a Reception Staff Member, I want to update patient appointment details so that records remain accurate.
- As a Reception Staff Member, I want to notify patients of schedule changes so that they stay informed.

5. Persona: Administrator

- As an Administrator, I want to manage user accounts so that authorized users can access the system securely.
- As an Administrator, I want to manage user roles and permissions so that users only access relevant information.
- As an Administrator, I want to monitor system activities so that security and compliance can be maintained.
- As an Administrator, I want to manage reviews and reported issues so that the quality of the system can be improved.

Product Backlog

ID	User Story	Priority
PBI-01	As an Administrator, I want to manage user accounts so that authorized users can access the system securely.	High
PBI-02	As an Administrator, I want to manage user roles and permissions so that users only access relevant information.	High
PBI-03	As a Patient, I want to register an account so that I can access the healthcare services online.	High
PBI-04	As a Reception Staff Member, I want to register new patients so that their information is available in the system.	High
PBI-05	As a Patient, I want to book and manage appointments so that I can consult doctors and dietitians conveniently.	High
PBI-06	As a Reception Staff Member, I want to manage appointments so that schedules remain organized and conflict-free.	High
PBI-07	As a Doctor, I want to access patient medical records so that I can make informed decisions during consultations.	High
PBI-08	As a Dietitian, I want to view patient health records so that I can provide accurate dietary advice.	High
PBI-09	As a Doctor, I want to record health check-up results so that patient information stays up to date.	High
PBI-10	As a Dietitian, I want to create personalized diet plans so that patients can achieve their health goals.	High

PBI-11	As a Reception Staff Member, I want to update patient appointment details so that records remain accurate.	Medium
PBI-12	As a Reception Staff Member, I want to notify patients of schedule changes so that they stay informed.	Medium
PBI-13	As a Doctor, I want to schedule follow-up consultations so that patients receive continuous care.	Medium
PBI-14	As a Patient, I want to view my health check-up results so that I can monitor my health condition.	Medium
PBI-15	As a Patient, I want to access my diet plan and progress records so that I can follow my nutritional goals effectively.	Medium
PBI-16	As a Dietitian, I want to update diet plans so that recommendations remain suitable for patient progress.	Medium
PBI-17	As a Doctor, I want to review patients' health progress so that I can recommend appropriate treatments.	Medium
PBI-18	As a Dietitian, I want to communicate with patients so that I can provide guidance and answer queries.	Medium
PBI-19	As an Administrator, I want to monitor system activities so that security and compliance can be maintained.	Low
PBI-20	As an Administrator, I want to manage reviews and reported issues so that the quality of the system can be improved.	Low
PBI-21	As an Operations Manager, I want to view reports on daily, weekly, and monthly patient visits so that I can understand how well the center is performing.	Low
PBI-22	As an Operations Manager, I want to generate a Monthly Business Performance Summary so that management has a quick overview of operations.	Low
PBI-23	As an Operations Manager, I want to track patient retention rates so that I can evaluate the organization's performance.	Low
PBI-24	As an Operations Manager, I want to review feedback and satisfaction summaries so that service quality can be improved.	Low

Sprint Plan & Backlog

Sprint 1: User Management & Registration

Focus: Establishing secure authentication, role mapping, and profile creation.

Task ID	Task Description	Est. Hours	Assigned Developer
T-1.1	Design backend architecture for secure user authentication.	8	IT25101803
T-1.2	Implement Administrator tools for managing roles and permissions.	6	IT25102636
T-1.3	Create UI/API for Patient account registration.	5	IT25103681
T-1.4	Create UI/API for Reception Staff to register new patients.	5	IT25101696
T-1.5	Execute testing on authentication and RBAC workflows.	4	IT25102636

Sprint 2: Appointment Management

Focus: Enabling conflict-free booking, schedule updates, and notifications.

Task ID	Task Description	Est. Hours	Assigned Developer
T-2.1	Design calendar interface for Patient booking.	6	IT25101803
T-2.2	Develop reception dashboard for schedule management.	7	IT25101803
T-2.3	Implement API for updating/canceling appointment records.	5	IT25102636
T-2.4	Integrate automated notifications for schedule changes.	5	IT25103681
T-2.5	Test booking flows and conflict prevention logic.	3	IT25101696

Sprint 3: Health Check-up Management

Focus: Doctor data entry, record access, and follow-up scheduling.

Task ID	Task Description	Est. Hours	Assigned Developer
T-3.1	Design Doctor interface for accessing medical records.	6	IT25102636
T-3.2	Create database tables to record health check-up results.	7	IT25101803
T-3.3	Develop Patient view for accessing personal check-up results.	5	IT25101696
T-3.4	Implement logic for Doctors to schedule follow-up visits.	4	IT25103681
T-3.5	Test data retrieval and authorization for medical records.	4	IT25101803

Sprint 4: Diet Plan, Communication & Review Management

Focus: Nutritional planning, messaging, analytics, and issue management.

Task ID	Task Description	Est. Hours	Assigned Developer
T-4.1	Build Dietitian module to create and update diet plans.	7	IT25102636
T-4.2	Implement secure messaging system between Patients and Dietitians.	6	IT25102636
T-4.3	Develop Administrator tools to manage reviews and system activity logs.	5	IT25103681
T-4.4	Create Management reporting dashboards (visitation/retention).	6	IT25101696
T-4.5	Conduct end-to-end testing of the complete system workflow.	5	IT25101803